

# Value of OSS Adoption for Grid Operators

## Methodology:

- Grid operator's choice between (1) proprietary and (2) OSS solutions
- Derived from game-theoretic discrete choice framework

**Focal metric:** Net benefit of OSS adoption measured in monetary terms

- Linear combination of value sources and weights (modular)
- Avoided costs, reduced risk, increased flexibility, downstream value

## Key features:

- Defined over fixed operating period and known load
- Multi-stakeholder perspective: grid operator, regulator, downstream consumer
- Easily accommodates counterfactual analysis and uncertainty
- Customizable through flexible parameterization → highly extensible
- Open source!

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## Timeline:

| Year | Month | Phase                                            |
|------|-------|--------------------------------------------------|
| 2026 | 2     | Begin model development                          |
|      | 3     | Model structure - feedback and refinement        |
|      | 5     | Model parameterization - feedback and refinement |
|      | 7     | Report development                               |
|      | 9     | Report publication and model release             |

# Components of Benefit to Cost Analysis

**Net benefit-to-cost** is calculated as a **weighted** sum of

- **Benefit** components, **strategic Value (V)** and **Social benefits (S)**
- **Cost** components, **Total Cost of ownership (TCO)** and **Risk (R)**.

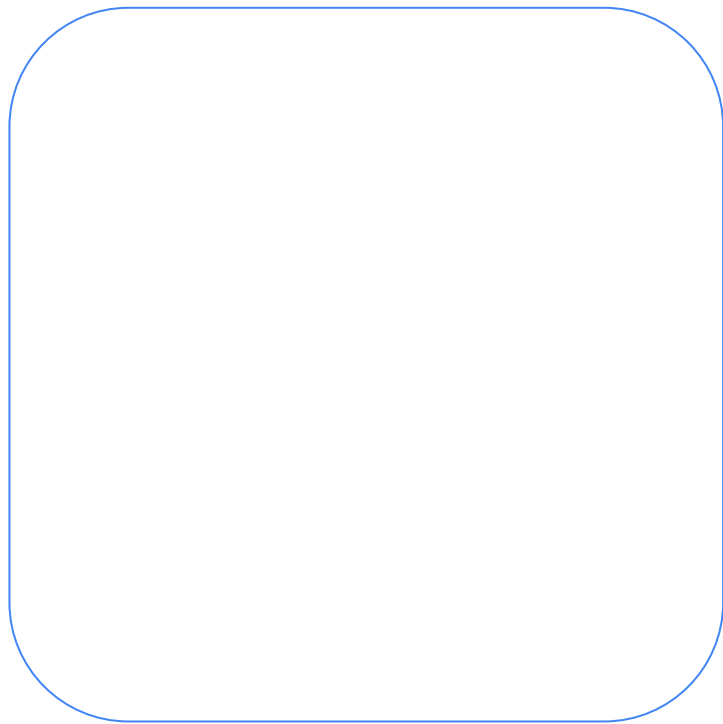
| Factor | Name                    | Type    | Description                                                                                                                                                                                                                                                                |
|--------|-------------------------|---------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| V      | Strategic value         | Benefit | Organization values, incl., Improvements to operational <b>efficiency</b> - reduced time-to-value i.e., <b>speed, innovations, Flexibility</b> (competitive procurement leverage), Multi-vendor <b>interoperability</b> /interchangeability, grid asset <b>utilization</b> |
| S      | Social benefits         | Benefit | Increased <b>Reliability/Resilience</b> ; environmental <b>sustainability</b> , and social <b>equity</b> , consumer/prosumer services- improved economic opportunity,                                                                                                      |
| TCO    | Total cost of ownership | Cost    | <b>CAPEX</b> : New specification design changes, qualifications, integration & commissioning<br><b>OPEX</b> : Operation & maintenance, upgrades, and support services, End-of-life                                                                                         |
| R      | Risk                    | Cost    | Service <b>reliability</b> risks, <b>regulatory</b> compliance, cyber <b>security</b> , and vendor <b>lock-in</b> risks                                                                                                                                                    |

$$B_o = W_V \Delta V + W_S \Delta S - W_O \Delta TCO - W_R \Delta R$$



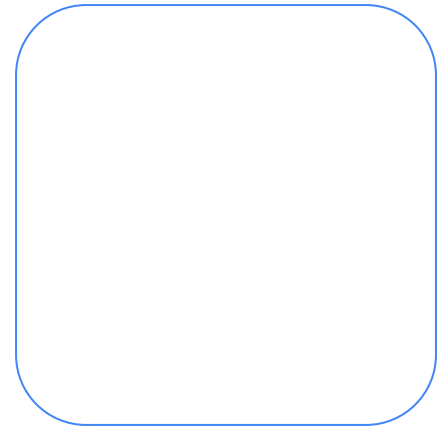
























| Time | Topic | Lead |
|------|-------|------|
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# Content Heavy Variants





